

Technical Architecture

Project design phase

Date	15 February 2025
Team ID	
Project Name	HouseHunt – House Rental Management System
Maximum Marks	4 marka

Solution Architecture

The HouseHunt solution architecture provides a scalable and secure web-based platform that connects tenants, landlords, and administrators through a centralized property rental system. The architecture follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

The main objectives of the architecture are:

- Provide a secure property rental platform.
- Enable role-based access for Tenants, Landlords, and Administrators.
- Manage authentication and authorization using JWT.
- Store property, booking, review, and user information securely in MongoDB.
- Provide RESTful APIs for seamless communication between the frontend and backend.
- Support scalable deployment and future enhancements.

Architecture Components

Layer	Components	Purpose
Presentation Layer	React.js, Vite, HTML, CSS, JavaScript	User Interface for Tenants, Landlords and Admin

Layer	Components	Purpose
Application Layer	Node.js, Express.js REST APIs	Business Logic and Request Processing
Authentication	JWT, BCrypt	Secure Login and Role-based Authorization
Database Layer	MongoDB with Mongoose	Stores Users, Properties, Bookings, Reviews and Wishlist
File Storage	Multer Uploads	Stores Property Images and Profile Images
Client Roles	Tenant, Landlord, Administrator	Different functionalities based on user role

System Workflow

1. User accesses the HouseHunt web application.
2. React frontend sends requests to Express REST APIs.
3. JWT Authentication verifies the user identity.
4. Business logic processes property listings, bookings, reviews, and wishlist operations.
5. MongoDB stores and retrieves application data.
6. Responses are returned to the frontend and displayed to the users.

Key Features

- User Registration and Login
- JWT Authentication
- Property Management
- Property Search
- Wishlist Management
- Booking Management
- Review System
- Admin Dashboard
- Role-Based Authorization
- Secure File Upload

- Responsive User Interface

Technologies Used

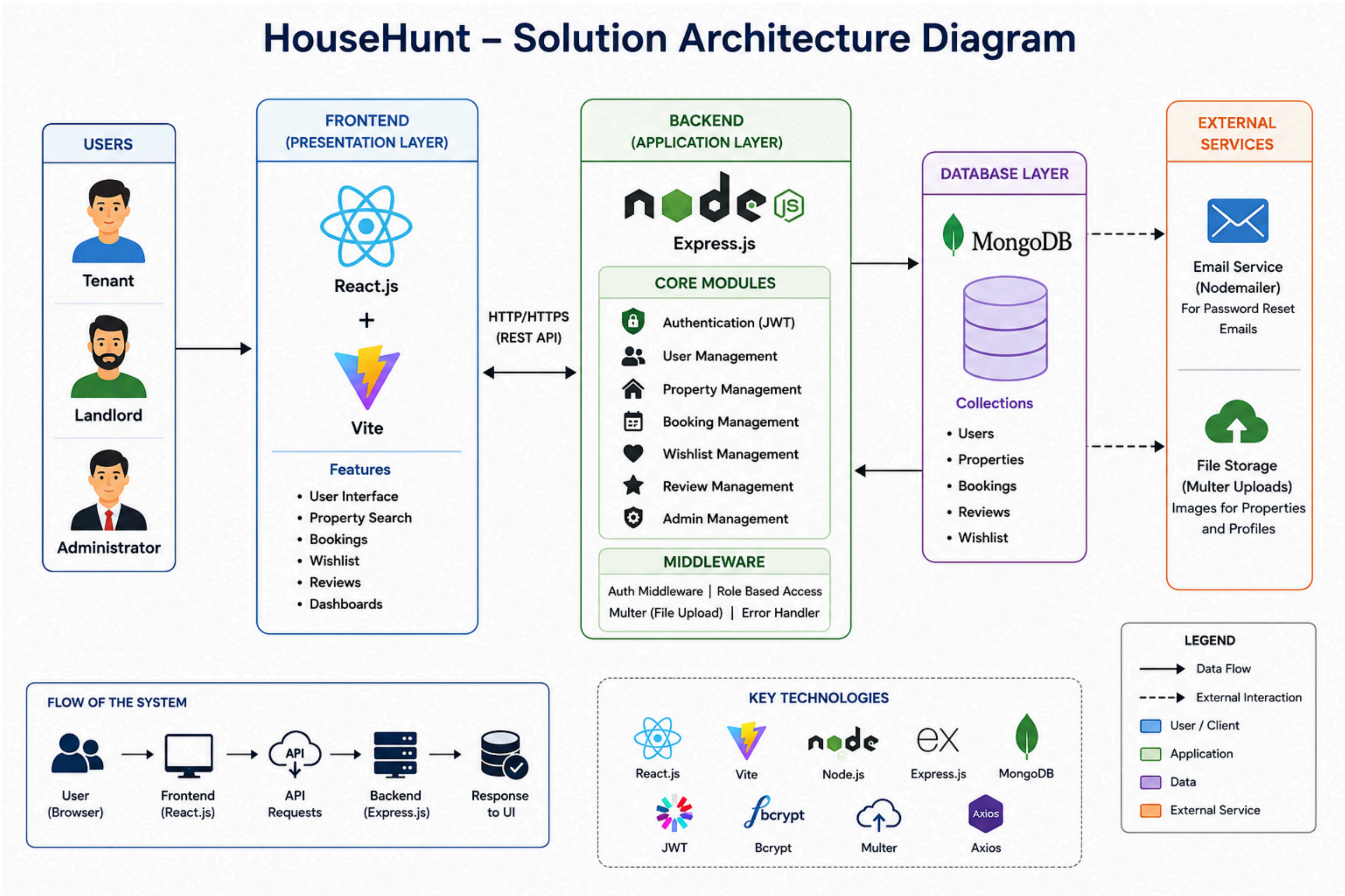
Technology	Purpose
React.js	Frontend
Vite	Frontend Build Tool
Node.js	Backend Runtime
Express.js	REST API
MongoDB	Database
Mongoose	ODM
JWT	Authentication
BCrypt	Password Encryption
Multer	Image Upload
Axios	API Communication

Benefits of the Architecture

- Modular Design
- High Scalability
- Secure Authentication

- Easy Maintenance
- RESTful Communication
- Efficient Database Management
- Responsive User Experience
- Future Ready for Cloud Deployment

2. Architecture Diagram



3. Architecture Description Table

Architecture Layer	Components	Description
Client Layer	Tenant, Landlord, Admin	Users access the application through a web browser

Presentation Layer	React.js, Vite	Displays the user interface and communicates with backend APIs
API Layer	Express.js REST API	Handles all client requests and responses
Authentication Layer	JWT, BCrypt	Provides secure authentication and authorization
Business Layer	Property, Booking, Wishlist, Review Modules	Executes application business logic
Database Layer	MongoDB, Mongoose	Stores all application data
File Storage	Multer	Manages image uploads for users and properties